

Program : Diploma in Tool & Die Engineering	
Course Code : 5119	Course Title: Advanced Machine Tools & Process Planning Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To impart knowledge in students about principles/methods of non-conventional machining in tooling.
- To provide skills on selection, operation and control of appropriate processes in tool/mould making.
- To prepare NC/CNC programs for various jobs.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge about conventional and non-conventional machining processes.		Production Engineering II	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the non-conventional machining processes in the making of Press tools & Moulds.	9	Applying
CO2	Demonstrate the characteristic features of ECM/EDM (Spark & Wire cut)	9	Applying
CO3	Prepare the part program, edit and execute on EDM (Spark & Wire cut)	15	Applying
CO 4	Illustrate the process planning layout for an industrial type progressive die/Injection Mould	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			3
CO2	3						
CO3	3			3			3
CO4	3					3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the non-conventional machining processes in the making of Press tools & Moulds.		
M1.01	Identify the major non- conventional machining like AJM, USM, ECM, ECG, CHM, EDM, EBM, LBM, IBM, PAM and the types of energies and methods of material removal.	3	Understanding
M1.02	Practice on reference point & NC Coordinate setting on EDM spark machine.	6	Applying
CO2	Demonstrate the Characteristic features of ECM/EDM (Spark & Wire cut)		
M2.01	Identify the parts and the operations, types of electrodes, die electric medium, selection of electrodes etc. of the EDM (Spark & Wire cut).	3	Understanding
M2.02	Practice on the feeding of wire electrode from the spool & reference point & NC Coordinate setting on EDM wire cut machine	6	Applying
	Lab Exam – I	1.5	
CO3	Prepare the part program, edit and execute on EDM (Spark & Wire cut)		
M3.01	Apply profile cutting on Core/ cavity Insert using spark EDM	9	Applying
M3.02	Apply profile cutting on a blanking die using wire cut EDM	6	Applying

CO4	Illustrate the process planning layout for an industrial type progressive die/Injection Mold.		
M4.01	Prepare process chart for each components of injection mold like sprue bush, cavity & core inserts, mold base, ejector pins etc.	6	Applying
M4.02	Prepare process chart for each components of press tool like die plates, stripper plate, punches etc.	3	Applying
	Lab Exam – II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	Manufacturing process for engineering materials – SeropeKalpakjian
R1	N.C. Machines and CAM - Pressman Williams
R2	A text book of Production Engineering - P.C. Sharma

Online Resources:

Sl.No	Website Link
1	http://www.crectirupati.com/sites/default/files/lecture_notes/MMM%20LECTURE%20NOTES%20FINAL.pdf
2	http://www.helmancnc.com/wire-edm-programming-example/
3	https://www.engineering.com/AdvancedManufacturing/ArticleID/10100/EDM-101-Electrical-Discharge-Machining-Basics.aspx